

Exploring Gen-AI applications in building research and industry: A review

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Abstract

This paper investigates the transformative potential of Generative AI (Gen-AI) technologies, particularly large language models, within the building industry. By leveraging these advanced AI tools, the study explores their application across key areas such as automated compliance checking and building design assistance. The research highlights how Gen-AI can automate labor-intensive processes, significantly improving efficiency and reducing costs in building practices. The paper first discusses the two widely applied fundamental models—Transformer and Diffusion model—and summarizes current pathways for accessing Gen-AI models and the most common techniques for customizing them. It then explores applications for text generation, such as compliance checking, control support, data mining, and building simulation input file editing. Additionally, it examines image generation, including direct generation through diffusion models and indirect generation through language model-supported template creation based on existing Computer-Aided Design or other design tools with rendering. The paper concludes with a comprehensive analysis of the current capabilities of Gen-AI in the building industry, outlining future directions for research and development, with the goal of paving the way for smarter, more effective, and responsive design, construction, and operational practices.

Keywords

generative AI
large language models (LLMs)
building industry
automatic compliance checking (ACC)
building design
retrieval-augmented generation (RAG)

Article History

Received: 2 January 2025

Revised: 20 March 2025

Accepted: 26 March 2025

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1 Introduction

U.S. buildings, which consume a substantial 38% of the nation's total energy usage (Valencia-Solares et al. 2023), present significant opportunity for efficiency improvements. Artificial Intelligence (AI) stands as a pivotal tool in transforming the building sector with better performance and higher efficiency (Roh et al. 2021), like machine learning in building load predictions (Zhang et al. 2021), reinforcement learning for building controls (Wang and Hong 2020), and deep learning support Computer-Aided Design (CAD) (Baduge et al. 2022). New colloquially known as generative artificial intelligence (Gen-AI), is a topic of much debate and one prominent application thus far is the production of high-quality artistic media for visual arts, concept art, music, and literature, as well as video and animation (Epstein et al. 2023).

The term Gen-AI refers to computational techniques that are capable of generating seemingly new, meaningful

content such as text, images, or audio from training data (Feuerriegel et al. 2024). Gen-AI has been successfully applied in areas like Code Copilot (Wermelinger 2023), translation (Pym and Hao 2024), and game computer graphics (Lee et al. 2023a), due to its ability to generate context-aware, high-quality outputs with minimal human input. Compared to traditional methods, it excels in automating complex tasks, improving accuracy, and significantly enhancing both speed and scalability in processes like coding, language translation, and graphical design. More important, it shows the capabilities to create new data sets (Feuerriegel et al. 2024). Thus, Gen-AI can also play a critical role in building industry, including building design (Li et al. 2024a), construction (Liao et al. 2024), and operation (Zhang and Chen 2023). The intricate mesh of design specifications, graphical data, and regulatory requirements provides a fertile ground for Gen-AI applications that can streamline processes, optimize material use, and ensure compliance with stringent safety and construction standards. However,

Nomenclature

ACC	automatic compliance checking	HVAC	heating, ventilation, and air conditioning
AEC	architecture, engineering, and construction	IDF	input data file
AI	artificial intelligence	IT	information technology
API	application programming interface	LLM	large language model
BEM	building energy modeling	LoRA	low-rank adaptation
BERT	bidirectional encoder representations from transformers	LSTM	long short-term memory
BIM	building information modeling	NLP	natural language processing
BTO	building technology office	NeRFs	neural radiance fields
CAD	computer-aided design	OCR	optical character recognition
CoT	chain-of-thought	PEFT	parameter-efficient fine-tuning
DOE	Department of Energy	PCS	physical and computational sciences
DQN	deep Q-network	RAG	retrieval-augmented generation
Gen-AI	generative artificial intelligence	SLM	small language model
GAN	generative adversarial network	UBEM	urban building energy modeling
GPT	generative pre-trained transformer	VAE	variational autoencoder

these challenges include the availability of robust datasets to train AI models, the accuracy and sensitivity of AI systems, and the alignment of these technologies with evolving energy efficiency assessments (Nah et al. 2023; Kenthapadi et al. 2023). This underscores the urgency and potential for integrating Gen-AI, refers to computational techniques that are capable of generating seemingly new, meaningful content (Feuerriegel et al. 2024), across building systems to catalyze advancements in energy management (Tomar and Grover 2024). Some studies summarize the applications of LLMs in the building energy sector (Xiao and Xu 2024; Liu et al. 2025), but there is a lack of review work that outlines the development of different Gen-AI models, explains key terminologies, and connects them to building applications.

This review begins with an overview of fundamental terminology and popular Gen-AI tools, analyzing their fundamental models (e.g., Transformer, Diffusion Model), functionalities (e.g., text and image generation), and accessibility (e.g., open-source, paid, API availability). Data were collected¹ from workshops, websites, newsletters, and other publicly available sources, focusing on model developers, fundamental architectures, application types, and access mechanisms. We then examine existing applications in the building industry and research, selecting studies through a systematic keyword search combining “Generative AI”, “Transformer”, “Large Language Model”, and “Diffusion Model” with “Building”, “Architecture”, and “Construction”. Using Web of Science and Google Scholar,

we included academic papers, reports, and preprints to capture emerging research. From 60 initially identified studies, we selected 11 representative works for further discussion. To structure the analysis, we apply an evaluation framework considering publication year, application area (e.g., compliance checking, building design, energy modeling, building control), application type (e.g., text- or image-based), model architecture, customization strategies, benefits over traditional methods, and current limitations. Finally, we discuss research priorities and emerging applications, highlighting key opportunities for Gen-AI integration in the building industry. The graphic abstract shows the outline of this paper.

The significance of this work lies in its comprehensive analysis of Gen-AI applications across multiple domains within the building industry and research, providing a holistic perspective that transcends previous studies focused on isolated aspects of AI integration. This research bridges the knowledge gap between AI technology and building-related practitioners by presenting accessible examples and technologies, ultimately enhancing our understanding of how Gen-AI can be further integrated into building industry practices and research.

2 Overview of current generative AI models

Numbers of Gen-AI tools are available nowadays. In this section, we will first introduce some basic terminologies and examine the available model tools and platforms. Then we will introduce the common way to customize the models.

¹ Data were initially collected in July 2024 and updated in January 2025.

2.1 Fundamental models

Fundamental models form the foundation of Gen-AI models, providing the core architectures and techniques that enable the generation of new data across various domains. The development of Gen-AI has been driven by the emergence of those key models. Variational Autoencoders (VAEs, 2013²) (Kingma and Welling 2022) introduced a probabilistic approach to data generation and reconstruction. Generative Adversarial Networks (GANs, 2014) (Goodfellow et al. 2014) employed a generator-discriminator framework to produce high-quality synthetic data, particularly in image generation. Diffusion Models (2015) (Sohl-Dickstein et al. 2015) leveraged iterative noise addition and removal to learn complex data distributions, advancing applications in image and audio synthesis. The Transformer architecture (2017) (Vaswani 2017) revolutionized natural language processing and later extended to multimodal tasks, forming the basis for modern large language models. Neural Radiance Fields (NeRFs, 2020) (Mildenhall et al. 2022) enabled photorealistic 3D scene reconstruction from 2D images by modeling light interactions. In this paper, we focus on transformer models, widely used for text generation (e.g., ChatGPT³), and diffusion models, commonly applied in image generation (e.g., Stable Diffusion⁴), as they are pivotal in cutting-edge building-related applications. Readers interested in other models can refer to the cited literature for further details.

2.1.1 Transformer model

Transformer is a type of neural network architecture (Wu and Feng 2018) that differs significantly from earlier recurrent neural architectures like long short-term memory (LSTM) (Yu et al. 2019). Unlike LSTMs that process data sequentially, transformers process entire sequences of data simultaneously (Zeyer et al. 2019). This parallel processing capability reduces training time, making transformers particularly efficient for handling large datasets. This efficiency has led to their widespread adoption in training models on extensive text corpora such as the Wikipedia database and the Common Crawl dataset (Hochreiter and Schmidhuber 1997). Developing a machine learning model, including those based on transformer models, is typically an iterative process where a data scientist might experiment with dozens to hundreds of model versions to achieve a model that satisfies predefined criteria (Vartak et al. 2016).

Referencing Figure 1, Transformer consists of two main components: encoders and decoders. The encoder captures the essence of the input data, such as a sentence in English, using a process called self-attention that helps it understand how each word relates to others. This processed information is then passed to the decoder, which also uses self-attention to maintain awareness of its own outputs to ensure coherence and relevance in its tasks, such as generating a sentence in Chinese. Additionally, the decoder uses outputs from the encoder to accurately piece together the final result, ensuring that each part of the output is a logical continuation of the input. This dual mechanism of understanding and generating data makes the Transformer exceptionally effective for tasks involving sequences of data.

A **Large Language Model (LLM)** is a type of Gen-AI model, built upon Transformer architecture, specifically designed to handle a variety of natural language processing (NLP) tasks, including text generation, language understanding, and classification (Wang et al. 2019), like Generative Pre-Trained Transformer 4 omni (GPT-4o) (Islam and Moushi 2024). As a widely-used example of Gen-AI, LLMs are known for their ability to generate human-like text, making them integral to applications such as chatbots, text-to-image tools, and other AI-driven content creation technologies (Radford et al. 2019). While LLMs offer promising enhancements in efficiency and decision-making within the building sector, their deployment must be carefully managed to address challenges related to data quality, security, and domain-specific requirements (Yao et al. 2024).

A Small Language Model (SLM) is a more resource-efficient alternative to LLMs, designed with significantly fewer parameters. SLMs excel in efficiency, requiring less computational power and producing a lower carbon footprint, making them more accessible to researchers and developers

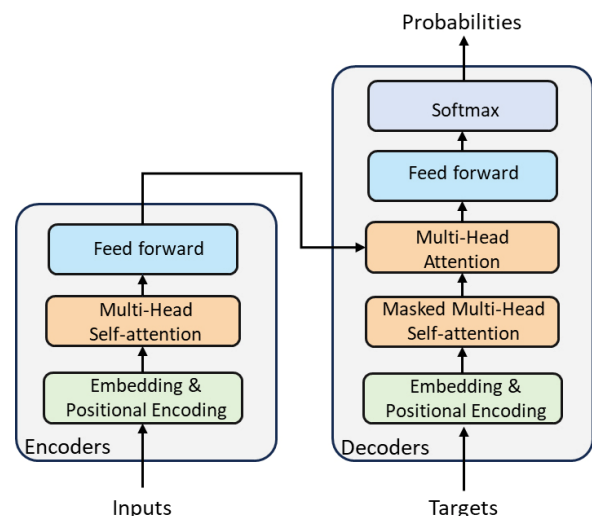


Fig. 1 Simplified Transformer model architecture

² The years represent the birthdates of the fundamental models.

³ Chat Generative Pre-Trained Transformer (ChatGPT): one of the most famous AI tools from Open AI (<https://chatgpt.com/>).

⁴ Stable Diffusion: one of the most famous text-to-image models (<https://stability.ai/>).

with limited resources (Schick and Schütze 2021) and can even be able to run local on smart devices (Xu et al. 2024). Typically, SLM can be distilled from an LLM (Jazbec et al. 2024). However, SLM tend to underperform in complex tasks compared to LLMs, which can process longer sequences and capture intricate contextual relationships more effectively (Schick and Schütze 2021). While SLMs offer a balance between performance and efficiency, they may fall short in scenarios requiring deep language understanding or few-shot learning.

Transformer models (or LLMs) are widely used for text generation (see Section 2.2.3 of this paper), and we will explore their applications in buildings in Section 3.1.

2.1.2 Diffusion model

Diffusion models are a class of generative models that create data, such as images or videos, by simulating a process that gradually transforms noise into meaningful content (Luo 2022), like Stable Diffusion and DALL-E (Borji 2022).

In the Diffusion model, as illustrated by Figure 2, the process begins with a clear and simple image—such as a sketch of a house—and gradually introduces controlled noise, distorting the image progressively. This model is adept at a unique process: it systematically learns to reverse this noise addition. By training on a multitude of examples, it acquires the ability to reconstruct clear images from these increasingly noisy states. The model meticulously removes the noise in a stepwise fashion, refining and enhancing the image until it reaches a high-quality final version. This technique showcases how diffusion models are employed to create or enhance images with precision and creativity, transforming basic outlines into detailed and coherent visuals.

As a fundamental framework in Gen-AI, Diffusion model operates via this two-step process, where the forward diffusion transforms data into noise and the reverse process, guided by a neural network, reconstructs the data from noise (Cao et al 2024). These models have gained prominence in generating high-quality images by refining random noise into detailed and coherent outputs, surpassing traditional models like GANs and VAEs in stability and quality. However, diffusion models often require longer sampling times due to their iterative denoising process, making them computationally intensive. Despite this, their ability to produce high-quality results across multiple domains, including image synthesis, text generation, and audio processing, makes them a powerful and versatile tool in Gen-AI.

Diffusion models are commonly used for image and video generation (see Sections 2.2.1 and 2.2.2 for further discussion). In building applications, diffusion models are used for direct generation, such as building floor plan design or envelope design (see Section 3.2.1 for more details).

2.2 Current models and platforms

In recent years, significant advancements in AI have given rise to highly sophisticated models capable of generating images, videos, and text. These models, primarily developed by leading Information Technology (IT) companies, leverage the latest in diffusion and transformer technologies to produce remarkable results. This section provides an overview of the most notable models in these domains, their availability, and the various ways they can be utilized. Table 1 sums part of the models and platforms⁵.

There are several ways to access and utilize these advanced AI models, which can be broadly grouped into open and non-open solutions. Open-source models, such as Meta's LLaMA, can be downloaded and deployed locally, offering users full control over their AI capabilities. On the other hand, non-open platforms like AWS Bedrock and Azure Portal provide hosted environments where users can interact with models without needing to manage the underlying infrastructure.

Utilization can also be categorized based on the application type (or user-interface type). For instance, AI models can be integrated into chatbot development, as seen with GPT-based chatbots, or used within web portals like the Azure Portal, which allows for real-time interaction and model testing. Additionally, tools like GitHub Copilot leverage AI for code generation and assistance, while API access, such as through the OpenAI API, enables developers to seamlessly incorporate AI functionalities into their applications and workflows. These diverse options allow users to choose the best approach for their specific needs, whether they prioritize control, ease of use, or integration capabilities.

2.2.1 Image generation models

Image generation models have seen substantial improvements, with models like OpenAI's DALL-E 2 (<https://openai.com/index/dall-e-2/>), Google's Imagen 3 (Google DeepMind 2024a), and Microsoft's NUWA-Infinity (<https://nuwa-infinity.microsoft.com/#/>) leading the pack. DALL-E 2, for example, can create highly detailed and realistic images from textual descriptions, making it a powerful tool for artists and designers. Similarly, Google's Imagen 3 excels in generating photorealistic images with fine details and complex compositions. These models are available through online playgrounds where users can experiment with generating images by inputting descriptive text. Additionally,

⁵ The information in Table 1 was collected in July 2024, and updated in March 2025. As website addresses may vary or be subject to change, the links are not all listed here. Readers interested in further details are encouraged to search the relevant terms using a search engine.

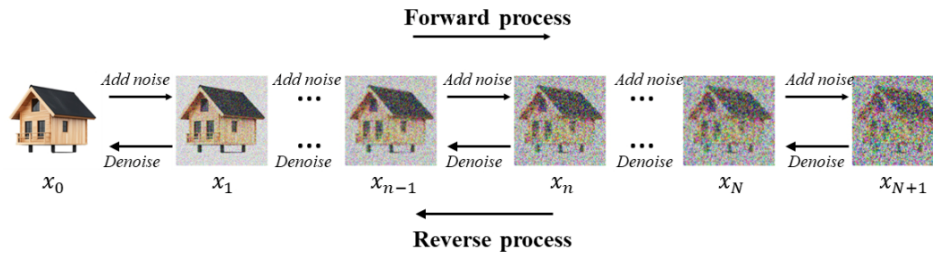


Fig. 2 Mechanism diagram of Diffusion model

Table 1 Some current available models and platforms

Company	Image generation model	Video generation model	Text generation model	Platform
OpenAI	DALL-E	Sora	GPT	OpenAI API Portal
Google	Imagen, Palm	Veo	Gemini*	Google DeepMind
Microsoft	NUWA-Infinity	NUWA-XL	Turing-NLG	Azure
Meta	Make-A-Scene	N/A	LLaMA*	N/A
Nvidia	Edify	N/A	Megatron-Turing NLG	Nvidia Playground
Tencent	GauGAN	N/A	Hunyuan	N/A
Alibaba	M6	N/A	AliceMind, Qwen*	Alibaba Cloud
Baidu	ERNIE-ViLG	N/A	ERNIE	Baidu Cloud
Others	Stable Diffusion, Midjourney, Runway, etc.	Runway, Pixverse, Phenaki, MagicVideo, etc.	Anthropic Claude, DeepSeek*, Grok, etc.	Hugging Face*, Amazon SageMaker etc.

*Open-source models or community

APIs (OpenAI Platform, <https://platform.openai.com>) provided by these companies allow developers to integrate image generation capabilities into their applications.

2.2.2 Video generation models

Video generation models, although still emerging, are showing great potential. Google's Veo (Google DeepMind 2024b) and Microsoft's NUWA-XL are at the forefront, capable of creating high-quality videos from text prompts. These models can be used to generate content for entertainment, education, and marketing purposes. Like image generation models, video generation models are accessible through APIs, enabling seamless integration into various digital platforms. Additionally, platforms like Runway (<https://runwayml.com/>) offer user-friendly interfaces where creators can experiment with video generation without needing deep technical knowledge.

2.2.3 Text generation models

Text generation models have advanced significantly, with prominent examples including OpenAI's GPT-4 (<https://openai.com/index/gpt-4/>), Google's Gemini (<https://gemini.google.com>), and Meta's LLaMA (<https://llama.meta.com/>). These models leverage Transformer architectures to produce coherent and contextually relevant text.

2.3 Common techniques for customizing models

In the field of Gen-AI, different techniques are employed, like prompt engineering, Retrieval-Augmented Generation (RAG), fine-tuning, architectural modifications, pretraining/Continual Pre-Training (CPT), etc. Some of the techniques can generally be ordered from low to high in terms of complexity, quality, cost, and time requirements as shown in Figure 3. However, it is important to note that this order is not absolute, as complexity and resource demands can vary depending on specific use cases. For example, full fine-tuning can, in certain scenarios, approach or even exceed the complexity and cost of CPT (Cossu et al. 2024). Additionally, there are numerous approaches to customizing models, and in this manuscript, we focus on highlighting some of the most well-known and widely used methods.

Furthermore, these methods are not mutually exclusive and are often combined to achieve optimal performance. For instance, CPT is frequently followed by fine-tuning to first adapt the model to a specific domain, after which it is optimized for a particular downstream task (Xia et al. 2024).

This paper will focus on detailing prompt engineering, RAG, and fine-tuning, as these techniques are the most widely utilized in current building research applications. Additionally, we will provide a brief overview of other common approaches, offering a reference point for future

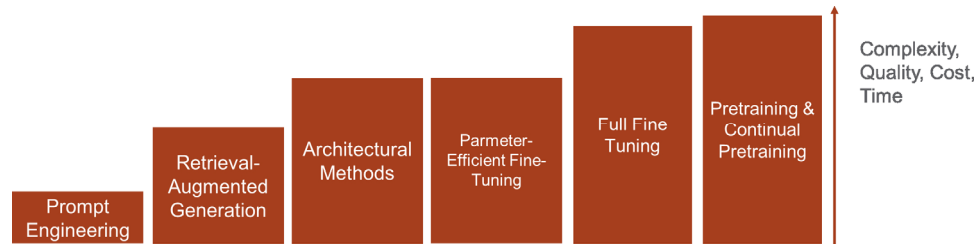


Fig. 3 Common techniques for customizing LLMs

researchers to explore and incorporate these methods into ongoing and future building-related studies.

2.3.1 Prompt engineering

Prompt engineering refers to the process of designing, crafting, and refining inputs or prompts to optimize the responses of Gen-AI models, which are LLMs based on NLP (Brown et al. 2020; Giray 2023). The quality and structure of prompts are critical in guiding these models to produce meaningful, accurate, and contextually relevant outputs. This makes prompt engineering crucial, as it influences how the AI interprets tasks, ensures clarity in communication, and helps guide the AI toward desired outcomes (Bozkurt and Sharma 2023). Prompt engineering helps in co-creation of content with AI, fostering deeper learning and more personalized interactions between AI and users. In the context of buildings, different prompts can affect the performance of updating building energy model files. We will discuss this further in Section 3.1.

Chain-of-Thought (CoT) is a newer, advanced prompting technique that guides models to generate intermediate reasoning steps, fostering more structured and logical problem-solving (Zhang et al. 2022). This approach is particularly effective for complex tasks such as mathematical problem-solving, commonsense reasoning, and multi-step decision-making due to its ability to break down problems into manageable, sequential steps (Wei et al. 2023). By promoting step-by-step thought processes, CoT enhances a model's ability to produce accurate, coherent, and contextually sound outputs, significantly improving performance on tasks that require deeper reasoning and reducing the model's reliance on surface-level pattern recognition (Wei et al. 2023). ChatGPT's new deep research function (Fortson 2025) is built upon this principle, structuring responses by first separating questions and then presenting detailed, logical answers. Although CoT has been seldom reported in current building-related research, it holds potential for future applications due to its capabilities in facilitating complex reasoning and calculations.

2.3.2 Retrieval-augmented generation

Gen-AI systems encounter challenges when processing

extended chat histories due to input length limitations, often leading to degraded performance as contextual information accumulates (Currie 2023). This issue is exacerbated by the high computational and temporal costs of handling lengthy messages. To overcome these constraints, RAG provides a structured framework that enhances text generation by dynamically incorporating external knowledge (Lewis et al. 2020). The mechanism of RAG involves three sequential stages. First, during the retrieval phase, the system queries a pre-indexed knowledge base—such as documents, databases, or domain-specific repositories—to identify and extract contextually relevant information. Next, in the integration phase, the retrieved data is fused with the original user input to create an enriched context. Finally, in the generation phase, the language model leverages this augmented input to produce outputs that are both accurate and contextually aligned. By supplementing the model's static pre-trained knowledge with real-time or domain-specific information, RAG addresses limitations in relevance and factual accuracy, particularly in dynamic or specialized domains (Fan et al. 2024).

Underpinning RAG's efficacy are embedding models, which serve as the backbone of its retrieval capability (Guu et al. 2020). These models convert discrete data, such as text tokens or images, into continuous vector representations that encapsulate semantic relationships. For instance, words with similar meanings or contextual associations are mapped to vectors that reside in proximate regions of the embedding space. When a user query is processed, its vector representation is compared against embeddings of indexed content to retrieve the most semantically relevant information. This process enables RAG to efficiently filter and prioritize external data, even when dealing with long or complex inputs, by reducing reliance on raw text processing and instead leveraging compact, numerically encoded representations.

RAG's adaptability makes it equally valuable for both SLMs and LLMs, empowering them to deliver precise, timely responses in scenarios requiring up-to-date or specialized knowledge (Fan et al. 2024). Implementation strategies vary in complexity, as demonstrated by the pipeline configurations in Table 2 (Gao et al. 2024). For deeper insights into these configurations, readers are encouraged to consult.

Table 2 Comparison between three RAG paradigms (Gao et al. 2024)

Pipeline	Module	Sub-module/process	Description
Naive RAG	Indexing	Basic indexing	Organizes data chunks into a retrievable form
	Retrieval	Standard retrieval	Retrieves documents based on query-document similarity
	Generation	Basic generation	Generates responses based on retrieved chunks
Advanced RAG	Indexing	Hierarchical indexing	Structures information for efficient retrieval
	Pre-retrieval	Query rewrite	Rewrites queries to improve retrieval precision
		HyDE	Hypothetical document generation to enhance retrieval accuracy
	Retrieval	Dense retriever	Retrieves documents using dense vectors for better semantics
	Post-retrieval	Rerank and filter	Prioritizes key information, filtering out noise
	Generation	Advanced generation	Utilizes reranked chunks to generate high-quality responses
Modular RAG	Indexing	Chunk optimization	Refines chunk size and overlap to balance context and noise
		Structure organization	Organizes documents hierarchically for better retrieval
	Pre-retrieval	Query expansion	Expands queries to cover more retrieval contexts
		Query transformation	Transforms original queries for more accurate results
	Retrieval	Retriever FT	Fine-tuning retrievers for task-specific performance
		Retriever source	Sources of retrieval include dense/sparse vectors and graphs
		Retriever selection	Selects between dense, sparse, and hybrid retrievers
	Post-retrieval	Compression	Compresses retrieved chunks to retain key information
		Selection	Directly removes irrelevant chunks
	Generation	Generator FT	Fine-tunes generators for task-specific output quality
		Verification	Verifies generated responses using external knowledge or models
	Routing	Pipeline routing	Routes queries to specific RAG pipelines based on content
	Scheduling	Recursive or iterative retrieval	Schedules repeated retrieval/generation cycles until completion
	Knowledge guide	Knowledge graph-driven reasoning	Guides retrieval using knowledge graphs for enhanced answers

RAG content indexing within the device allows AI models to efficiently search through a predefined set of documents or knowledge bases, further reducing dependency on cloud-based retrieval services. This ability to perform RAG locally strengthens the system's response speed and privacy control, as data remains on the device.

In building applications, RAG is especially useful for tasks such as Automatic Compliance Checking (ACC). It can store various data types, such as standard files and model data, as vectors in a vector database. When information is needed, a search is performed within the vector database, retrieving only the relevant data based on similarity scores. This retrieved data is then mapped back to the original files and passed to the LLMs. Further discussion can be found in Section 3.1.1.

2.3.3 Fine-tuning

Fine-tuning is a technique used to adapt a pre-trained model⁶ to a specific task or dataset by further training it on a

smaller, more focused dataset (Ohm 2024). This approach is highly effective because it leverages the general knowledge acquired by the model during its initial training on large and diverse datasets, while refining its capabilities to excel in particular areas of interest.

2.3.3.1 Full fine-tuning

Full fine-tuning updates all parameters of a pre-trained model, allowing it to deeply specialize for a target task. A seminal example is Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al. 2019), which demonstrated how bidirectional transformer models could be fine-tuned for diverse NLP tasks like question answering and sentiment analysis. For instance, a general-purpose LLM like GPT-4 can be fully fine-tuned for applications such as legal document analysis or medical text generation by adjusting its weights to align with task-specific data. This method typically involves training the model for fewer epochs with a reduced learning rate to prevent overfitting, preserving its broad knowledge while enhancing specialized performance. However, full fine-tuning requires substantial computational resources, making it less practical for scenarios with limited hardware or frequent task updates.

⁶ Pre-Training is the initial phase of training large language models on vast, diverse datasets (e.g., books, web text) to build foundational language understanding and general-purpose capabilities.

2.3.3.2 Parameter-efficient fine-tuning (PEFT)

Parameter-efficient fine-tuning (PEFT) methods update only a small subset of parameters, achieving a balance between performance and computational efficiency. One such approach involves adapter layers, which insert lightweight modules between transformer layers, allowing only these components to be tuned while the base model remains frozen (Houlsby et al. 2019). Another technique, LoRA (Low-Rank Adaptation), injects trainable low-rank matrices into attention layers, approximating full weight updates with minimal parameter adjustments (Hu et al. 2022). Similarly, prefix-tuning optimizes continuous “soft prompts”—virtual tokens prepended to inputs—to steer model behavior without modifying its underlying architecture (Li and Liang 2021). These methods achieve performance comparable to full fine-tuning while training less than 5% of the model’s parameters. For instance, LoRA fine-tunes GPT-3 for dialogue tasks with only ~0.1% of its parameters updated, significantly reducing memory and computational costs (Hu et al. 2022).

In addition to full fine-tuning and parameter-efficient methods, LLMs can be tailored through objective-driven approaches like instruction tuning (enhancing task generalization via formatted instructions (Zhang et al. 2024b)), alignment tuning (ensuring ethical compliance via human feedback (Dai et al. 2023)), etc. These strategies prioritize specific behavioral or functional goals rather than purely optimizing computational efficiency. For a comprehensive review of these and other adaptation paradigms, readers may refer to foundational surveys such as Zhao et al. (2023).

Fine-tuning is also a key technology widely used in the building industry to customize existing models for optimized performance in specific areas, such as generating and updating building energy modeling files and scripts. Further details can be found in Section 3.1.3.

2.3.4 Other techniques

2.3.4.1 Architectural methods

Architectural methods focus on optimizing model design and performance to improve scalability, speed, and resource utilization (Frantar and Alistarh 2023). These include distillation and compression, which reduce model size while preserving functionality. For instance, DistilBERT employs knowledge distillation, where a smaller student model is trained to mimic the outputs of a larger teacher model (e.g., BERT) using techniques like soft label matching and layer reduction, achieving 40% fewer parameters with minimal performance loss (Sanh et al. 2020). Compression methods like quantization reduce numerical precision (e.g., from 32-bit to 8-bit) to accelerate inference and decrease memory

usage, while pruning eliminates redundant weights to shrink model size. Sparse attention further optimizes the Transformer architecture by limiting token interactions to critical dependencies, enabling efficient processing of longer sequences (Frantar and Alistarh 2023). Together, these approaches balance performance with practical deployment constraints.

2.3.4.2 Continual pre-training

The CPT approach involves extended pre-training on a domain or task, followed by supervised training on the downstream task (Uppaal et al. 2024). CPT expands a model’s general knowledge by further training on large, domain-specific data, improving its overall performance in a domain, while fine-tuning adapts a model for a specific task using a smaller, labeled dataset, optimizing it for that particular application (Parmar et al. 2024). For instance, if a consulting firm pre-trained a model on general architectural texts, they may later want it to analyze energy efficiency regulations for green building certifications. Initially, the model may struggle with specialized terminology. Through CPT, training on datasets like LEED guidelines and energy efficiency standards, the model can better understand sustainability-focused language. However, to specifically identify violations in energy compliance reports, the company would apply fine-tuning on a labeled dataset of compliance reports and violations, optimizing the model for this precise task without further broadening its overall domain knowledge.

Recent advances also integrate hybrid systems like combining RAG with parameter-efficient fine-tuning to balance dynamic knowledge retrieval and domain-specific adaptation (Sarmah et al. 2024). Tool-augmented frameworks like Toolformer (Schick et al. 2023) and Gorilla (Patil et al. 2024) enable LLMs to self-supervise API calls, dynamically accessing external tools (calculators, databases) for real-time data or specialized computations. These approaches synergize parametric knowledge with external interfaces, enhancing robustness in open-world tasks while minimizing retraining overhead. Engaging readers can check the reference for more details.

2.4 Summary

The current landscape of AI models for image, video, and text generation is shaped by innovations from leading technology companies, leveraging advanced core technologies such as diffusion models and transformer models. Diffusion models are particularly effective in generating highly detailed and realistic images, refining visual outputs through iterative processes. In contrast, transformer models excel

at understanding and generating sequential data, making them ideal for text and video generation tasks.

Many of the latest AI models combine these two technologies, capitalizing on their complementary strengths to enhance performance and versatility across various applications. These models are accessible through a range of platforms and methods. For instance, users can interact with these models in real-time through online playgrounds, integrate AI capabilities into existing applications via APIs, or deploy models locally using downloadable open-source projects.

Additionally, techniques such as fine-tuning and RAG are becoming increasingly important for customizing these models to specific tasks and environments. Fine-tuning adapts pre-trained models for specialized applications by refining them with targeted data, while embedding models allow AI capabilities to be integrated directly into applications and devices for real-time processing.

3 Generative AI for building research and industry

This section of the manuscript delves into the application of Gen-AI in building research and industry, focusing specifically on two prevalent types of applications: text generation and image generation, as shown in Figure 4. These models represent cutting-edge approaches in the field of Gen-AI, offering unique advantages in processing and generating complex data sets pertinent to building design, construction, and management. By exploring the capabilities and distinctions of these models, we aim to elucidate how they can be effectively utilized to address various challenges in the building industry, enhancing both the efficiency and precision of predictive analytics and decision-making processes.

3.1 Text generation

Text generation—particularly for indexing, version tracking, and summarizing documentation—is one of the most

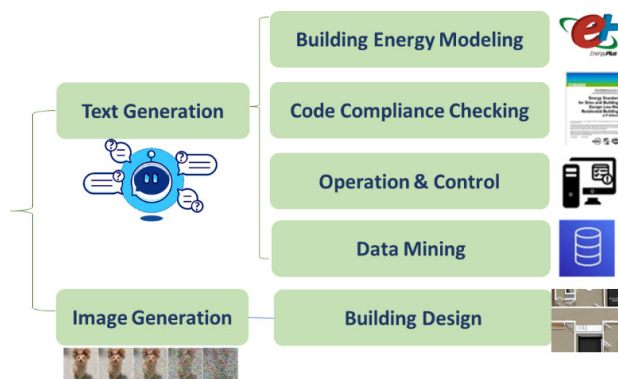


Fig. 4 Application for Gen AI in building industry and research

straightforward applications of LLMs. Liu et al. identified the areas such as automated energy model generation, energy management optimization, fault detection and diagnosis, data-driven forecasting, and building design and retrofit support in building energy system (Liu et al. 2025).

3.1.1 Automatic compliance checking

3.1.1.1 NLP-based automatic compliance checking

Using NLP for ACC process is not new. Lee et al. (2023b) presented a method to improve ACC by defining high-level, implementable methods that enhance low-level rule-checking procedures, as shown in Figure 5, thereby automating the translation of natural language building regulations into explicitly defined rules.

Lee et al. (2023b) summarised the issues for the traditional NLP process ambiguity in interpretation, cultural and regional variations, necessity for human intervention, cross-referencing issues, and challenges in computer processing.

3.1.1.2 GPT-based automatic compliance checking

However, those conclusions are based on the birth before LLM become popular. In fact, most of the issues can be handled using the novel LLM techniques.

Liu et al. (2023) focused on developing an ACC for building design specifications by leveraging deep learning and LLMs. Figure 6 shows the detailed workflow. Their study involved evaluating GPT-based models through four distinct tests to determine their effectiveness in comprehending and processing regulatory documents. To enhance accuracy, they fine-tuned their own model specifically trained on building specifications. Their fine-tuned GPT-3 model can achieve 91% accuracy as the best record compared with other OpenAI models. They also found some drawbacks: the models cannot remember the content of several segmented but related tasks; computing capacity for large batch processing is restricted; the current models cannot understand the information in higher dimensions which are frequently applied in projects including sounds, images, information models, etc.

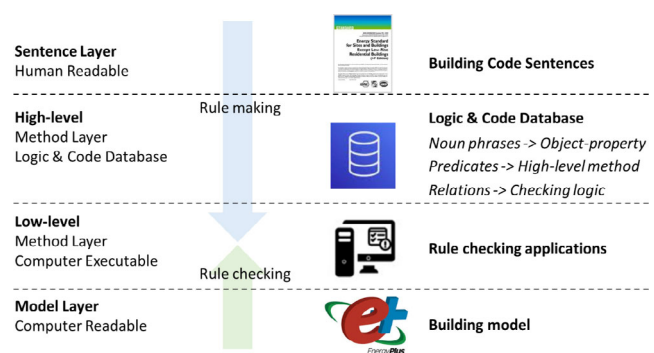


Fig. 5 Logic rule-based mechanism for ACC (redrawn based on information from Lee et al. (2023b))



Fig. 6 Design of Liu et al.'s implementation (redrawn based on information from Liu et al. (2023))

Chen et al. (2024) introduced a framework integrating LLMs, deep learning, and ontology knowledge models. In their case studies, the framework effectively automates the extraction and interpretation of complex regulatory texts from the documents, converting them into structured formats using a combination of deep learning classification models (Text Convolutional Neural Network, LSTM, and BERT) and GPT-4 for structured information extraction. The paper identifies limitations, such as challenges in processing qualitative regulations related to aesthetics or spatial functionality, and the need for further optimization to handle more complex building systems and regulations.

3.1.1.3 A case study for automatic compliance checking under Azure-OpenAI flow works

The authors also leverage a LLM for building documents ACC as a case study, with the primary purpose of using LLMs for content indexing and RAG. This approach enables the extraction of detailed requirements from building standard documents such as ASHRAE 90.1 (Goel et al. 2014) and International Energy Conservation Code (IECC) (Mapes and Conover 2012). For example, it can precisely extract specific R-value requirements for a particular climate zone and file version, such as ASHRAE 90.1 2019. The system can leverage two distinct workflows to achieve these tasks effectively, as demonstrated in Table 3: one operates online through the Azure OpenAI API (Azure OpenAI 2024), utilizing GPT-4 and embedding models without fine-tuning, while the other employs a local implementation using Hugging Face with the LLaMA model. Only open-source files are tested using workflow 1 (Azure workflow). For the Azure workflow, specifically for the ACC tasks, deploying the embedding model incurs a cost of approximately \$0.10 per code file book, while one round

of QA costs around \$0.01. In contrast, the Hugging Face workflow is free, as it runs locally on our infrastructure.

Figure 7 illustrates our customized RAG process designed specifically for building ACC. The process begins with input parsing, where various documents such as building plans, energy models, and standard files are extracted and processed. These inputs are then transformed into vector representations using an embedding model and stored in a vector database, which supports efficient retrieval of relevant sections. An initial prompt is generated based on the parsed inputs and retrieved data, which is then processed by a LLM via an API to produce detailed responses. A caching mechanism is employed to store frequently accessed results, enhancing the efficiency of the process. The generated outputs are filtered to ensure relevance and accuracy before being presented through the application's user interface, allowing users such as building inspectors and designers to review the compliance results and take necessary actions. This tailored RAG approach ensures thorough and efficient evaluation of building designs against requirements.

We also developed a local User Interface (Chatbot) using Workflow 2 as shown by Figure 8.

To test our demo chatbot, we developed a series of scenarios, as shown in Appendix I Table A1.

As a result, we found, most of the issues bring out by the previous researchers can be handled: by providing the specific standard files, the novel tool can check the requirement based on the users' inputs (like WA state tested in the demo); a chatbot can support the user interact with the models and a copilot stand by with the file maybe a future direction for this development; by using embedding models and divided the files into different sections and search using similarity functions, the cross-referencing issue can be directly handled. Using RAG process can also handle the problem that Liu et al. (2023) mentioned for large data handle with limitation of memories. Novel GPT-4 and other advanced model can also handle complicated inputs, like tables, images, and voice data (Eagalapati et al. 2024). Those studies underscore the practical relevance of Gen-AI in the building industry, offering scalable solutions that reduce labor costs, enhance regulatory compliance accuracy, and support more efficient project management.

Table 3 Work flows

	Workflow 1	Workflow 2
Embedding Model	Text-embedding-ada-002	Text-embedding-inference
GPT	Gpt-4 1106-Preview	LLaMA 3.3
Platform	Azure OpenAI Service	Hugging Face
Cost	Count based on tokens	Free

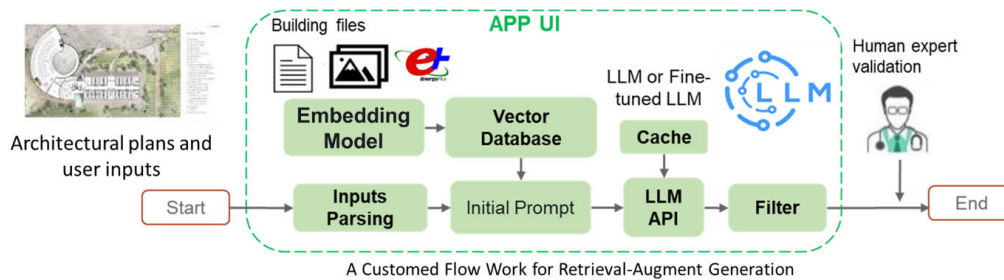


Fig. 7 Flowchart for customized RAG for building ACC

Fig. 8 Demo local chatbot for building ACC (Workflow 2)

3.1.2 Data mining

LLMs can be used to generate content in a given format, making them useful for data mining. Zhang et al. (2024c) presented an automated data mining framework that combines maximal frequent itemset mining with GPT to identify energy waste patterns in building operational data, as shown in Figure 9. The framework improves the efficiency of extracting valuable operation patterns by reducing redundancy and transforming these patterns into prompts for GPT. This process automates the analysis of large datasets, liberating users from tedious manual evaluations. The framework was tested on a year's worth of data from a real-world building chiller plant system, successfully detecting various energy waste patterns, such as valve faults and improper device coordination, with a detection accuracy of 89.17% for energy waste patterns and 99.48% for normal operations. The study highlights the cost-effectiveness and competitive

edge of using GPT for this purpose, with a total cost of \$17.68 (compared to several weeks of effort typically required by an engineer), demonstrating its potential to replace human analysts in identifying and addressing energy inefficiencies in building systems. Their framework can be extended to the analysis of other building systems data, where maximal frequent itemset mining can be used to reduce noise, and the resulting patterns can be further analyzed.

In a more recent study, Xiao and Xu (2024) developed an LLM-based multi-agent framework to automate building energy optimization tasks by effectively handling unstructured building data. The framework operates in three key stages: building information processing, performance diagnosis, and retrofit recommendation. Each stage leverages LLM agents, enhanced with domain-specific knowledge through RAG, to process diverse and unstructured inputs like energy audit reports. The multi-agent system assigns specific roles—planner, researcher, and advisor—to facilitate task automation, from extracting building metadata to diagnosing energy performance and generating retrofit suggestions. Case studies demonstrate the framework's ability to reduce manual labor, deliver reliable outputs, and handle varied data inputs with a cost-effective performance. This approach bridges data-driven and knowledge-driven methods, providing a scalable, trustworthy solution for building energy data management.

3.1.3 Building energy modeling support

LLMs also work for text-like file generation, like JSON scripts and Python scripts (Porter and Zingaro 2024). In building industry, EnergyPlus (Crawley et al. 2000) Input Data File (IDF) files are also commonly used for building energy modeling (BEM) software (Mendes et al. 2024), and IDF's generation and editing can benefit from LLMs. This approach is very different from how traditional machine learning approaches support energy modeling (Liu et al. 2024).

Zhang et al. (2024a) and Zhang and Chen (2023) detailed three case studies for LLMs in BEM: the first case study focuses on IDF's generation, reducing the manual effort required for model setup and error correction; the second

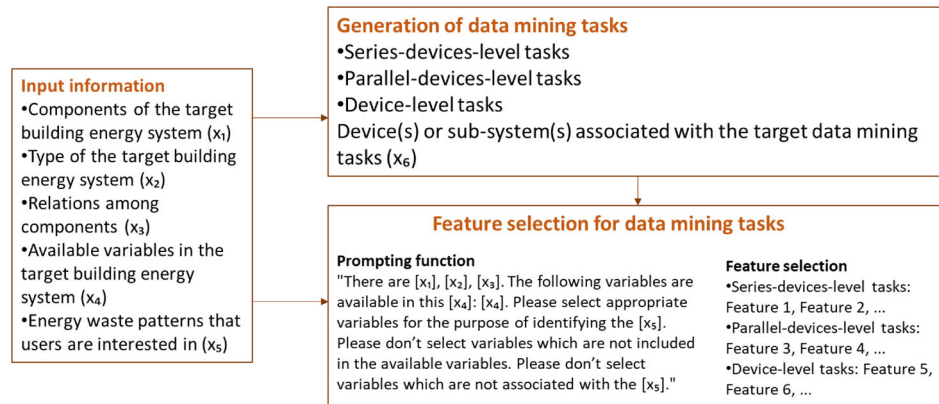


Fig. 9 Illustration of kernel density estimation-based transformation (redrawn based on the information from Zhang et al. (2024c))

case study highlights simulation output visualization, demonstrating LLMs' capabilities in generating Python scripts for data analysis and producing professional-grade visualizations with minimal human intervention; the third case study leverages RAG to enhance simulation knowledge extraction and training, using a dynamic, AI-driven interaction with resources like BEMyclopedia to provide users with educational materials and practical guidance. The paper also discusses the challenges associated with LLMs, including significant computational demands and potential consistency issues, but notes that ongoing technological advancements are addressing these limitations. The findings advocate for a multidisciplinary approach, combining expertise from AI and building modeling to effectively utilize LLMs in BEM. They recommended future research should focus on creating specialized LLMs, such as "BEMGPT", tailored specifically for BEM to further sustainable building solutions.

Jiang et al. (2024) introduce Eplus-LLM, an innovative

platform that leverages a fine-tuned LLM specifically Flan-T5 (Bergner et al. 2024), to automate BEM by translating natural language descriptions directly into EnergyPlus simulation files. Figure 10 shows the framework. The platform addresses critical challenges in building design and analysis, such as the need for specialized knowledge, complex software proficiency, and time-intensive modeling processes. By incorporating four distinct prompt formats, Eplus-LLM flexibly handles varied building geometries, occupancy scenarios, and equipment loads, reducing manual modeling efforts by over 95% and achieving 100% accuracy across 152 validation cases. Its robustness is demonstrated through resilience to varied user inputs, including different tones, misspellings, omissions, and redundancies, ensuring adaptability in real-world applications. The system tokenized and embodied natural language inputs, enabling the LLM to generate precise building models and simulation results through the EnergyPlus engine. Validation results highlight not only significant time efficiency but also alignment with

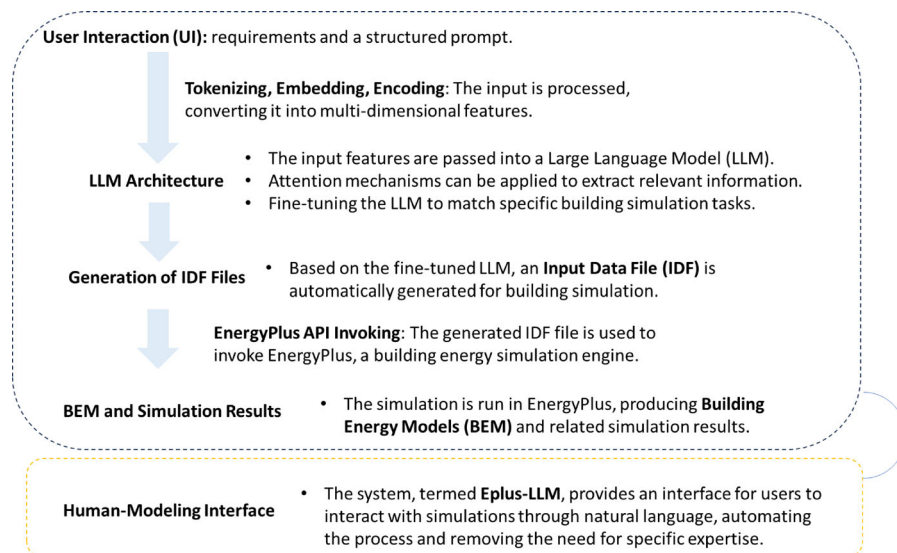


Fig. 10 The framework of the Eplus-LLM platform (redrawn based on the information from Jiang et al. (2024))

expert manual modeling in terms of accuracy and output quality. However, Eplus-LLM currently handles relatively simple modeling scenarios and faces limitations in processing complex geometries, multi-zone configurations, and interdependent parameters due to GPU constraints and LLM performance. These findings underscore the transformative role of Gen-AI in optimizing building modeling processes, paving the way for more scalable, accessible, and efficient simulation practices while highlighting areas for future enhancement to broaden real-world applicability.

Jiang et al.'s recent study (Jiang et al. 2025) focused on prompt engineering, highlighting how it uncovers efficiencies and optimize modeling processes. Three case studies of varying complexity were conducted: simple rectangular building modeling, complex structures with varied window-to-wall ratios and internal loads, and real-world applications involving detailed geometry and customizable materials. The findings indicate that prompt engineering, particularly one-shot and few-shot learning strategies, effectively automates BEM tasks, with compact LLMs like Llama3 and Mistral-v0.3 outperforming larger models despite their smaller size and shorter context windows. One-shot learning yielded "Perfect" results in simpler tasks, while complex scenarios required multiple demonstrations to achieve similar outcomes. Although zero-shot learning consistently failed due to the absence of prior domain knowledge, chain-of-thought strategies offered moderate improvements, especially when tasks were divided into manageable sub-tasks. The study underscores Gen-AI's ability to significantly reduce manual effort, streamline urban-scale modeling, and support retrofit and design optimizations, while also identifying limitations such as difficulties in handling highly complex geometries and the sensitivity of model performance to prompt design.

As a summary, LLMs shows significant potential in advancing efficient BEM by automating and optimizing complex tasks such as simulation input generation, output visualization, error analysis, and co-simulation. Case studies demonstrate that techniques like prompt engineering, multi-agent systems, and RAG can streamline BEM workflows, reduce manual effort, and enhance accessibility for non-experts. However, challenges remain in handling intricate zoning, complex geometries, and ensuring model consistency. Future research should focus on refining LLM performance through advanced prompt engineering, developing domain-specific models and integrating multi-modal AI to support visual modeling and semantic interpretation. Collaborative efforts between AI researchers and building energy experts are essential to bridge knowledge gaps, improve model adaptability, and fully leverage LLMs' potential for large-scale energy modeling (Mondal et al. 2024) and intelligent building management (Fan et al.

2023). These advancements will not only support energy efficiency but also broaden the application of Generative AI throughout the building lifecycle, enhancing its practical relevance in real-world practices.

3.1.4 Control support

Since LLMs are pre-trained models, they incorporate some common knowledge and calculation capabilities, making them useful for supporting robot control (Wake et al. 2023; Ye et al. 2023). Researchers have also explored their applications in building control.

Ahn et al. (2023) investigated the use of artificial intelligence, specifically ChatGPT and Deep Q-Network (DQN) models (a reinforcement learning approach), to optimize the control of HVAC systems in a reference office building simulated using EnergyPlus (Crawley et al. 2001) as shown in Figure 11. The goal was to minimize energy consumption while maintaining indoor CO₂ levels below 1000 ppm. ChatGPT, utilizing its pre-trained language model capabilities via the OpenAI Chat Completion API, provided real-time decision-making without traditional training processes. In contrast, the DQN underwent 500 iterations over an 11-day period to refine its policy. Over a 3-day evaluation period, the ChatGPT control achieved a 16.8% reduction in energy use, whereas the DQN achieved a 24.1% reduction. Although the DQN showed higher energy savings, ChatGPT demonstrated significant potential by leveraging real-time building operational data and pre-learned domain knowledge. Future research will focus on fine-tuning ChatGPT for more specialized HVAC control tasks and managing complex building state information. The study highlights the potential of generative AI in autonomous building system operations, showcasing its capability to provide plausible control actions, such as adjusting outdoor damper openings and target leaving chilled water temperatures, to enhance energy efficiency.

3.2 Image generation

Using GANs model or VAE model support 2D or 3D architecture design is not new. De Miguel Rodríguez et al. (2020) used VAEs for generating preliminary 3D forms from input parameters. Karadag et al. (2023) and Tanasra et al. (2023) utilized GANs model to refine architectural floor plans, obtaining plans with flat furniture layouts. Engaging readers can refer to (Li et al. 2024b) for reference. In this section, we will only focus on how diffusion model and transformer model may assist for building related image generation.

Currently, there are two main approaches for building design image generation: one involves direct generation using diffusion models followed by manual design selection;

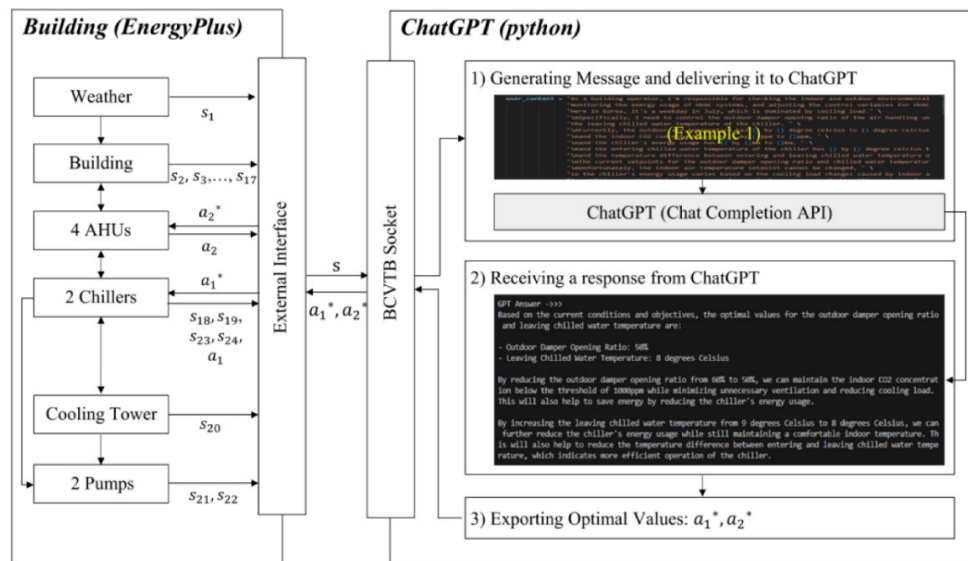


Fig. 11 Co-simulation of EnergyPlus and ChatGPT (Ahn et al. 2023)

the other employs an indirect approach using LLMs or other text generation models to create structured templates (typically in JSON format) that can be interpreted by rendering tools such as CAD software.

3.2.1 Direct generation

Direct generation typically involves sending prompts to diffusion models, which then generate a list of candidate figures. Jo et al. (2024) proposed a novel approach and tool, named AIBIM-Design, for generating photorealistic building façade images using Gen-AI, as shown in Figure 12. The approach addresses the significant time and resource

demands of existing tools in architectural visualization, which are essential throughout a project's lifecycle for communicating designs to stakeholders and facilitating design improvements. Leveraging generative AI, the proposed method offers a more efficient and scalable solution, capable of producing a broader range of design alternatives. The feasibility of this approach is demonstrated through intensive testing within a specific design context, focusing on generating façade design alternatives that reflect local identities. The study confirmed that including region-specific prompts in the AI model leads to higher quality and more detailed images. To achieve this, the model was trained using a dataset paired

Trigger words (common prompts: building facade, storefront image)	None	Korea style	Tokyo style	Manhattan style
Output images				

Fig. 12 Images generated by entering various scale-specific building façade design style prompts (redrawn based on the information from Jo et al. (2024))

with text data, obtained and preprocessed from street-view images of different regions. The trained model was validated with neighborhood facilities in Seoul, successfully transforming mass geometry into photorealistic images.

The research resulted in the generation of approximately 2000 façade images, consuming significant but manageable hardware resources. The AIBIM-Design tool, developed from this study, is accessible to both architectural experts and the public, allowing the generation of real-life images that represent diverse design identities from mass-type models. Future directions include expanding the training models horizontally to cover various building façade types and categories, and vertically to incorporate different kinds of visualizations beyond façades. This generative AI-based visualization tool demonstrates the potential to enhance financial and temporal efficiency in the design communication phase of architectural projects, providing a user-friendly interface for generating design alternatives that resonate with the local identity of the area.

Li et al. (2024a) explored the application of generative AI techniques in the generation of architectural floor plans, emphasizing advancements that allow architects to incorporate more conditional constraints into the design process, as shown in Figure 13. Architects can provide text data, such as client design requirements and architectural standards, and image data, including site plans and floor layout diagrams, to generative models. These inputs enable models to produce detailed and contextually relevant floor plans. The paper highlights the use of “Scene Graphs”, which describe the elements and their interrelations within a scene, making them suitable for depicting architectural floor plans. Technologies like SceneGenie integrate diffusion models to accurately generate these plans using Scene Graphs. Moreover, advanced generative models such as Stable Diffusion and Imagen can refine the generation process through text prompts and layout controls, producing comprehensive architectural designs based on input text. Despite these advancements,

the generated images often lack professional standards and rational layout adherence. However, the incorporation of additional constraints, like bounding boxes, shows promise for better alignment with architectural design considerations. The paper concludes that integrating professional architectural data with computational data, as illustrated by layout and segmentation masks, enhances the generative process, making it more efficient and capable of producing high-quality architectural floor plans.

3.2.2 Indirect generation

Indirect generation, on the other hand, uses language models to generate an input data file for a visualization tool (usually CAD software), which is then used to generate figures. Ko et al. (2023) proposed an innovative framework that incorporates ChatGPT and Veras, alongside parametric modeling and BIM to revolutionize the design process. This integration facilitates rapid design ideation and exploration, enabling architects to input design intentions via natural language, which ChatGPT translates into actionable modeling commands in software environments like Dynamo for Revit (Dynamo BIM, <https://dynamobim.org/>). Figure 14 shows their flow work. This not only enhances design efficiency and creativity but also encourages a more iterative and collaborative process. Veras contributes by generating realistic design visualizations, providing immediate feedback that guides further refinements. Despite these advancements, the framework is not without challenges. It requires precise and clear language inputs to produce accurate results, involves complex setup and integration with existing systems, and relies heavily on the designers' capability to iterate and refine AI-generated outputs, which may not always adhere to practical construction standards or architectural feasibility.

Other commercial companies are also involved in this area, leveraging their existing resources in BIM and data management. One notable example involves using Rhino and Grasshopper scripts (DaniGA 2024). By utilizing the



Fig. 13 Existing generative models can generate layout diagrams of rooms based on input text and can also be controlled accordingly based on input layouts (redrawn based on the information from Li et al. (2024a))

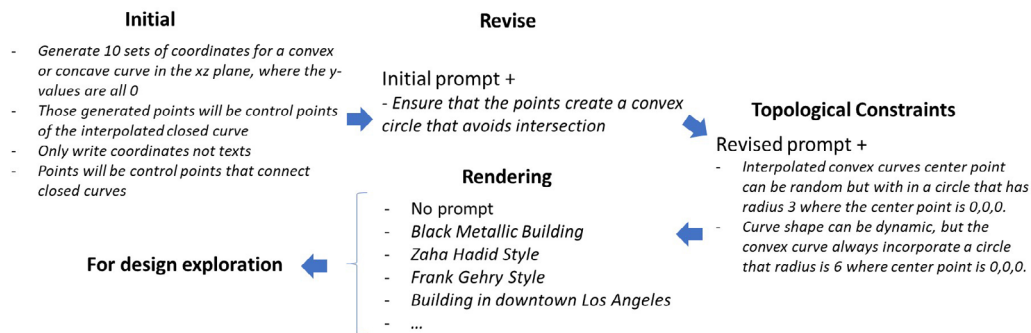


Fig. 14 Design exploration process through ChatGPT and Veras

OpenAI API, these companies assist in updating scripts in Grasshopper. Then, using their existing visualization tools, they can generate building designs.

3.3 Summary

Table 4 summarizes the current the typical studies using Gen-AI tools in building research and industry. The systematic review of recent studies reveals several notable trends and limitations in the application of Gen-AI within the building industry. A significant observation is the predominant reliance on ChatGPT models (both 3.5 and 4.0 versions) for text generation tasks across multiple applications. While these models have demonstrated effectiveness in various contexts, such as ACC, BEM updates, and control logic explanation, the narrow focus on ChatGPT models indicates a gap in exploring alternative LLMs. Particularly, the potential benefits of open-source models (like LLaMA from Meta), which could offer advantages in terms of cost-effectiveness and data privacy, remain largely unexplored. Furthermore, the current implementations tend to be case-specific solutions rather than generalizable frameworks, limiting their broader applicability across different scenarios in the building industry. This suggests a need for more comprehensive investigations that consider a wider range of model architectures and develop more versatile solutions applicable across diverse building industry contexts. Additionally, nearly all studies focus on general prompt engineering to customize models, with some attention given to RAG and fine-tuned models; however, few studies report the use of other approaches, such as CoT. Therefore, further research is needed, particularly to determine whether any customized approaches quantitatively improve performance. In image generation, direct generation using diffusion models faces challenges in control and precision, often failing to produce exactly what the user needs. As a result, significant manual effort is required to select appropriate results from the candidates. On the other hand, indirect generation approaches can only generate figures within the constraints of existing visualization software, limiting their

ability to create images beyond the software's capabilities. In the future, potential directions include improving control and guidance in direct generation, expanding the capabilities of indirect approaches, or integrating the two methods.

4 Discussion

This section addresses the needs on both the Gen-AI model development side and the application side. We discuss these needs and future work in three parts: gaps in current literature, needs identified in current application areas, and potential new application areas that could benefit from Gen-AI.

4.1 Gaps in current research

4.1.1 Model development

Our review highlights an over-reliance on the ChatGPT family of models for text generation, with limited exploration of alternative models. Customization techniques also require further exploration beyond prompt engineering. Advanced methods like CoT for structured reasoning and CPT for domain-specific performance enhancement are underutilized. Robust performance evaluation and benchmarking are essential, with quantitative metrics needed to assess customization. Developing domain-specific benchmarks will ensure practical model outputs. Addressing computational demands is critical, with resource-efficient models like SLMs offering lower computational requirements and broader accessibility. Finally, interdisciplinary collaboration between AI researchers and building professionals is vital for refining models and ensuring practical relevance.

4.1.2 Application

In text generation tasks, models need to handle complex, multidimensional regulatory documents and diverse data types. BEM applications show promise in automating modeling tasks but struggle with intricate geometries and multi-zone configurations. Refining prompt engineering

Table 4 Typical studies using Gen-AI tools in building research and industry

Reference	Application	Type	Model	Customizing models	Highlights	Limitations
Ko et al. 2023	Building Design	Image	GPT-4	Prompt engineering	LLM support for user inputs; Veras supports rendering	Issues in bespoke design; overlooks physical feasibility
Ahn et al. 2023	Building Control	Text	GPT-4	Prompt engineering	Explains EnergyPlus control logic; suggests control settings	Poorer performance (50% less efficient) compared with reinforcement learning approach
Liu et al. 2023	ACC	Text	GPT-3.5	Fine-tuning	Fine-tuned model for specific training data	Memory limits for related tasks; restricted computing for large batches; struggles with multidimensional data
Authors' team 2024	ACC	Text	GPT-4, LLaMA	RAG, Prompt engineering	Design a Chatbot based on RAG for documentation processes	Specific to certain tasks
Chen et al. 2024	ACC	Text	GPT-4	Fine-tuning	Integrates LLM with deep learning for structured information extraction	Challenges handling complex texts with nested clauses and conditional statements
Qin et al. 2024	Building Design	Text	GPT-3.5	Prompt engineering	Translates design requirements; 30x speed increase	Specific to certain context
Jo et al. 2024	Building Design	Image	Stable Diffusion 2.1	Prompt engineering	Input textual words; studies trigger words	Specific design context; requires 2 days for 2000 images
Jiang et al. 2024	BEM Support	Text	T5	Fine-tuning	Translates natural language descriptions to EnergyPlus models	Requires high consistency in outputs, struggles with diverse language formats
Zhang et al. 2024a	BEM Support	Text	ChatGPT 4.0	Prompt engineering	Generates/modifies IDF; visualizes EnergyPlus outputs	Self-consistency issues
Xiao and Xu 2024	Data Mining	Text	GPT-4	RAG, Multi-agent framework	Multi-agent framework for energy optimization using unstructured data	Challenges with domain-specific knowledge and handling diverse data inputs
Zhang et al. 2024c	Building Control	Text	GPT-3.5, 4	Prompt engineering	Constructs prompt functions; merges different layers	Poor performance in feature selection and knowledge interpretation

techniques and integrating multimodal AI could expand applicability. LLM-based HVAC controls demonstrate potential but currently underperform compared to reinforcement learning models. Fine-tuning LLMs for specific control tasks and developing multi-agent systems could bridge this gap.

Direct generation using diffusion models faces challenges in control and precision, often requiring manual intervention. Improving guidance mechanisms could enhance accuracy and reduce post-processing efforts. Indirect generation via language models for CAD input is limited by current rendering tools. Expanding these tools or developing hybrid methods could offer more versatile solutions for building design.

4.2 Current needs in building research and industry

4.2.1 Image processing techniques

A pivotal area requiring advancement lies in image processing techniques. Current Optical Character Recognition (OCR)

technologies proficiently extract text from images, yet they struggle with the complex visual interpretation needed for floor plans, blueprints, and site layouts. The industry requires advanced image processing techniques to enable direct use of architectural design files for tasks like ACC and BEM with Gen-AI. Future research should concentrate on crafting advanced image processing tools capable of accurately interpreting such intricate details. Additionally, people already made efforts on transforming CAD models into BEM (Xiao et al.) to help building early design stage; integrating Gen-AI tools directly with these systems could further streamline the automation process, reducing the dependency on manual inspections and enhancing overall workflow efficiency.

4.2.2 Choosing the appropriate Gen-AI integration approach

Determining the best implementation method for Gen-AI in the building sector is essential. Options include online chatbots for straightforward tasks, API access for automating specific workflows, or developing integrated software solutions like a script copilot. Chatbots may be ideal for

quick queries and user interactions, while APIs can streamline complex processes like compliance checks or model updates.

4.2.3 Privacy and cybersecurity in system integration

Using online AI services poses privacy risks, as data may be used in future model training. For sensitive and confidential information, local deployment is preferred, though it requires significant resources and may result in performance trade-offs. Thus, strengthening privacy protocols and cybersecurity measures when integrating LLMs online is essential, and this includes implementing advanced encryption, role-based access controls, and defending against prompt injection attacks and other vulnerabilities.

4.2.4 Comparisons among different models and compared with human efforts

While it is observed that GPT-4.0 shows improved performance over its predecessor GPT-3.5, there is a significant deficiency in comparative research across different model families. This includes a lack of detailed comparisons between open-source and commercial models. Additionally, while Zhang et al. (2024c) have highlighted the cost-effectiveness of using GPT models for data mining, asserting a cost of only \$17.68 versus the several weeks of effort typically required by an engineer, similar quantitative comparisons for other tasks like ACC are notably absent. This highlights a pressing need for more comprehensive studies that not only compare different AI models but also evaluate the efficiency of these models against traditional human efforts in the building industry.

4.3 Possible other application areas

4.3.1 Technical assistance and workforce development

Beyond the immediate needs of the building industry, AI technologies have the potential to revolutionize technical assistance and workforce training across various fields. For instance, by employing diffusion models, people may generate high-quality educational videos with integrated audio instructions, offering a comprehensive and immersive learning experience. These AI-generated videos could be particularly useful in training scenarios where real-world simulations would be prohibitively expensive or dangerous, such as refrigerant charge procedures and the associated risks of leaks or fire hazards. Such AI-driven simulations allow trainees to safely engage with realistic scenarios, ensuring they are well-prepared for actual fieldwork.

4.3.2 Building industry script copilot

AI also has the potential to revolutionize the building industry

by serving as a “script copilot”. AI-powered assistants would help professionals efficiently navigate and comply with complex building regulations, including energy consumption, fire safety standards, and environmental guidelines. Beyond traditional script files, the copilot could also interpret and manage any software or file formats that rely on text-based structures, such as JSON, CSV, XML, and similar configurations. For instance, integrating AI with platforms like the Modelica Building Library (Wetter et al. 2014) could automate the validation of building designs against regulatory requirements. The AI would intelligently interpret relevant files and apply them to specific design elements, streamlining the compliance process, minimizing human error, and enhancing efficiency. This functionality would be particularly beneficial in large-scale projects where manual compliance checks are both time-consuming and susceptible to oversight.

5 Conclusions

The integration of Gen-AI into the building industry is poised to revolutionize design, compliance, and training aspects significantly. This review underscores the transformative potential of transformer models and diffusion models to streamline and enhance processes that are traditionally labor-intensive and prone to errors. By leveraging Gen-AI techniques for ACC, data mining, BEM support, control support, and building design optimization, the industry can achieve substantial improvements in efficiency, accuracy, and safety:

1. **Text generation:** Gen-AI, particularly through prompt engineering, RAG and fine-tuning techniques, has achieved up to 91% accuracy in ACC. LLMs significantly reduce manual effort in tasks like generating and modifying BEM IDF files and visualizing simulation outputs. However, challenges remain in processing multidimensional data and ensuring consistency across related tasks.
2. **Image generation:** Diffusion models and LLM-supported template creation have shown promise in generating architectural designs and visualizations, achieving high-quality outputs with reduced manual input. Direct generation approaches can produce thousands of design alternatives in days, while indirect methods provide controlled outputs for CAD tools. However, limitations in control precision and visualization software constraints remain.
3. **Future needs:** Despite promising advancements, there is an over-reliance on specific models like ChatGPT, with limited exploration of open-source alternatives. Quantitative comparisons across different models and

evaluations against traditional human efforts are lacking. Addressing computational demands, enhancing privacy protocols, and developing standardized benchmarks are crucial for future research. Additionally, expanding the use of resource-efficient models like SLMs and fostering interdisciplinary collaboration will be key to fully leveraging Gen-AI in the building industry.

In essence, Gen-AI represents a transformative force within the building industry, offering innovative pathways for improving design, compliance, and training. Embracing these technologies enables the industry to advance resilience, safety, and efficiency, setting the stage for continued progress in building practices and compliance adherence. The future of the building industry will undoubtedly be

shaped by the integration of Gen-AI, facilitating smarter, more responsive, and sustainable construction practices.

Appendix I Sample testing results for automatic compliance checking

The chatbot was assessed on its ability to identify and interpret key requirements. Some sample prompt testing results are shown in Table A1. Note, this test involves the development of a localized chatbot system based on a LLama model (work flow 2 as mentioned in the manuscript), intended exclusively for internal use at Pacific Northwest National Laboratory Building Energy Code Program team. The system is designed for testing research capabilities only

Table A1 Sample prompts testing

Loaded file	Input prompts	Outputs
ASHRAE Standard 90.1	“What is the R value requirement for Washington State for external wall for commercial buildings?”	“Washington State primarily falls under Climate Zone 4C, and according to ASHRAE Standard 90.1, the requirements are as follows: Mass walls: Minimum R-value: R-9.5 continuous insulation (c.i.) Maximum U-value: U-0.104 Metal building walls: Minimum R-value: R-0 + R-15.8 c.i. Maximum U-value: U-0.060 Steel-framed walls: Minimum R-value: R-13 + R-7.5 c.i. Maximum U-value: U-0.064 Wood-framed and other walls: Minimum R-value: R-13 + R-3.8 c.i. or R-20 Maximum U-value: U-0.064.”
ASHRAE Standard 90.1	“What are the requirements for economizers in air conditioning systems?”	“Air economizer design capacity: Air economizer systems must be capable of modulating outdoor air and return air dampers to provide up to 100% of the design supply air quantity as outdoor air for cooling. Control signal: Economizer controls must be capable of and configured to sequence the dampers with the mechanical cooling equipment. The control should not rely solely on mixed-air temperature, except in systems controlled from space temperature, such as single-zone systems. High-limit shutoff: All air economizers must automatically reduce outdoor air intake to the design minimum when outdoor air intake will no longer reduce cooling energy use. The specific high-limit shutoff control types and set points vary by climate zone and are detailed in Table 6.5.1.1.3. Dampers: Exhaust/relief and outdoor air dampers must meet specific sealing requirements, and return dampers must meet the requirements for motorized exhaust/relief dampers. Relief of excess outdoor air: Systems must provide a means to relieve excess outdoor air during air economizer operation to prevent overpressurizing the building. The relief air outlet should be located to avoid recirculation into the building.”

Table A1 Sample prompts testing (Continued)

Loaded file	Input prompts	Outputs
ASHRAE Standard 90.1	“What are the guidelines for solar-ready roofs?”	“Solar-ready zone: A designated solar-ready zone must be allocated on the roof or other suitable area of the building to accommodate future solar photovoltaic or solar thermal installations. This zone should be free of obstructions such as HVAC equipment, vents, or other fixtures.
		Solar-ready area: The solar-ready zone must meet specific size requirements, generally based on a percentage of the total roof area or a minimum required area. The exact area may vary depending on the building’s location, type, and other factors.
		Pathways for conduits: Pathways must be provided for routing conduit or piping from the solar-ready zone to the electrical service panel or hot water system. These pathways should be planned to minimize the length and complexity of future installations.
		Structural load: The roof structure should be designed to support the additional load of solar panels, including both dead loads and live loads. Structural engineers must consider these loads in the initial design phase to avoid future complications.
		Documentation: The construction documents must include details of the solar-ready zone, conduit pathways, and any structural modifications required to accommodate solar installations. This information should be easily accessible for future reference by building owners or contractors.”

and operates within a controlled environment. All outputs are subject to human review.

Acknowledgements

The authors would like to express their gratitude for the support of the U.S. Department of Energy’s Office of Energy Efficiency and Renewable Energy (EERE) through Battelle Memorial Institute under Contract No. DE-AC05-76RL01830. The authors also acknowledge Lingzhe Wang for his assistance with formatting and technical editing.

Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article. Jian Zhang is an Editorial Board member of *Building Simulation*.

Author contribution statement

Hanlong Wan: conceptualization, methodology, validation, formal analysis, software, writing—original draft, visualization, investigation. Jian Zhang: supervision, funding acquisition, writing—review & editing, project administration. Yan Chen: conceptualization—AI for FDD, writing—AI in building control, writing—review & editing. Weili Xu: conceptualization, writing—copilot application, writing—review & editing. Fan Feng: formal analysis—Workflow 2 setup for ACC, visualization—outline and conceptual figure.

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